CASE STUDY: "PLANT ESCAPE" EDUCATIONAL AR APP

Role:	Information Technology Developer
Tech Stack:	Unity 3D, Vuforia Engine (AR), C#, Blender 3D
Publication:	JATIKA Academic Journal (Vol. 3, No. 4, 2022)
Achievements:	Google Play Store Release & Two Granted IP Rights (HKI)

1. Project Overview

During the COVID-19 pandemic, distance learning caused a significant "Learning Loss" among middle school students, particularly in subjects requiring laboratory observations like Biology (Plant Structures). Funded through the PKM program, my cross-functional team developed **Plant Escape**—an interactive Science Card Game coupled with an Augmented Reality (AR) mobile application. The physical cards act as markers, which, when scanned by the app, project 3D interactive biological models, bridging the gap between physical and digital learning.

2. Key Engineering Challenges & Solutions

A. Reliable Augmented Reality Marker Tracking

The Challenge: Designing an educational game requires high usability. The AR camera needed to consistently recognize and track physical "Science Cards" under varying lighting conditions and distances (from classrooms to outdoors), without jittering the 3D models.

The Solution: I integrated the Vuforia Engine with Unity 3D, heavily optimizing the image targets. By calibrating the high-contrast markers and testing the camera limits through Blackbox testing, the application achieved a 100% successful render rate across dynamic scan distances ranging from 5cm to 100cm.

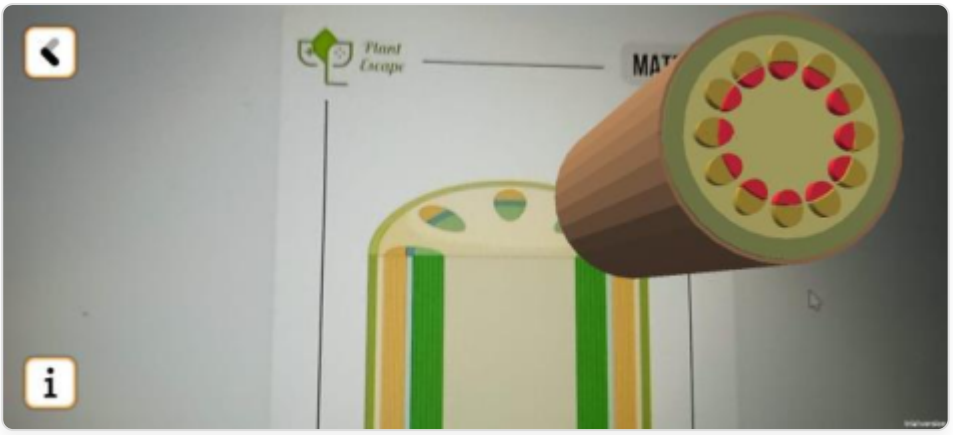
B. End-to-End Product Delivery & Hardware Optimization

The Challenge: The target demographic was public middle school students (SMP), meaning the application had to run smoothly on lower-end Android devices without crashing due to heavy 3D asset rendering.

The Solution: Employing the Multimedia Development Life Cycle (MDLC), I managed the asset pipeline. The 3D models (imported via fbx/gITF formats) were heavily compressed. Ultimately, the product was highly successful: it significantly improved student N-Gain scores in testing, was successfully published to the

Google Play Store, and earned two official Intellectual Property Rights (Hak Kekayaan Intelektual) from the Indonesian Government.

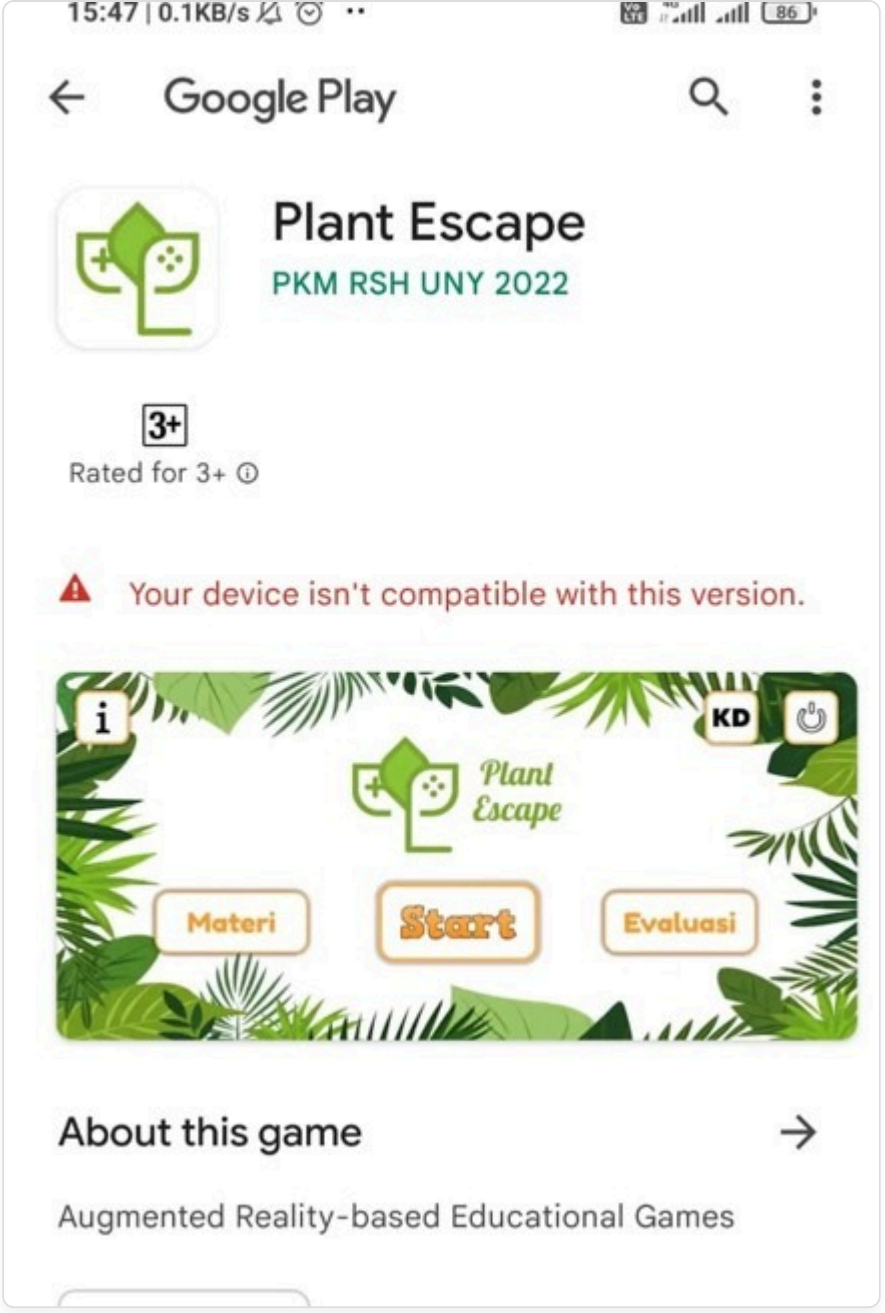
3. Product Demonstration & Output



Augmented Reality Marker Tracking: Scanning physical Science Cards to render interactive 3D biological models via Vuforia Engine.



The physical "Plant Escape" Science Card Game set distributed to middle schools.



End-to-End Delivery: Successfully published to the Google Play Store.